

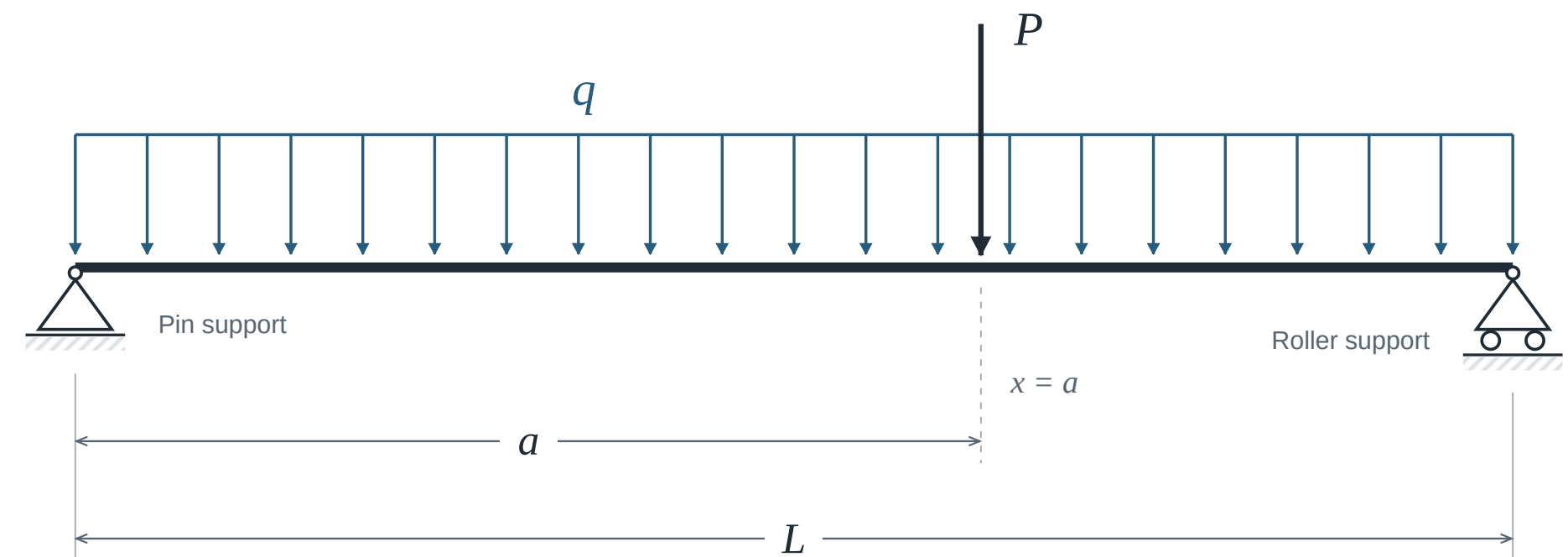
Machine Learning Project

Euler–Bernoulli Beam FEM and Neural-Network Surrogate

Physics-guided prediction of deflection, bending moment and shear force

Morris Wang
B.S. in Civil Engineering
National Chung Hsing University

github.com/morris2003-alt/beam-fem-neural-network-surrogate



Beam Model and Dataset

Model	Specification
Geometry and material	$L = 6\text{ m}$, $E = 200\text{ GPa}$
Rectangular section	$b = 0.15\text{ m}$, $h = 0.30\text{ m}$
Discretization	40 two-node cubic Hermite elements, 41 nodes
Loading ranges	$q = 2\text{--}15\text{ kN/m}$, $P = 5\text{--}45\text{ kN}$, $a/L = 0.15\text{--}0.85$

350 complete load cases

245 training / 53 validation / 52 test cases
All spatial samples from a case stay in one partition.

Reference responses

FEM supplies nodal deflection. Section equilibrium supplies moment and shear, including the point-load jump.

Physics-guided surrogate design

Output transformations enforce zero deflection and moment at the supports.
Separate shear networks represent the discontinuity at the point load.

Response	Representation	Hidden layers
Deflection	$\mathbf{W}_{\text{pred}} = \mathbf{W}_{\text{scale}} \times \xi(1 - \xi) \times \mathbf{Nw}$	64 / 64 / 32
Bending moment	$\mathbf{M}_{\text{pred}} = \mathbf{M}_{\text{scale}} \times \xi(1 - \xi) \times \mathbf{Nm}$	48 / 48 / 24
Shear force	Separate networks for $x < a$ and $x \geq a$	32 / 32 each

Inputs and training

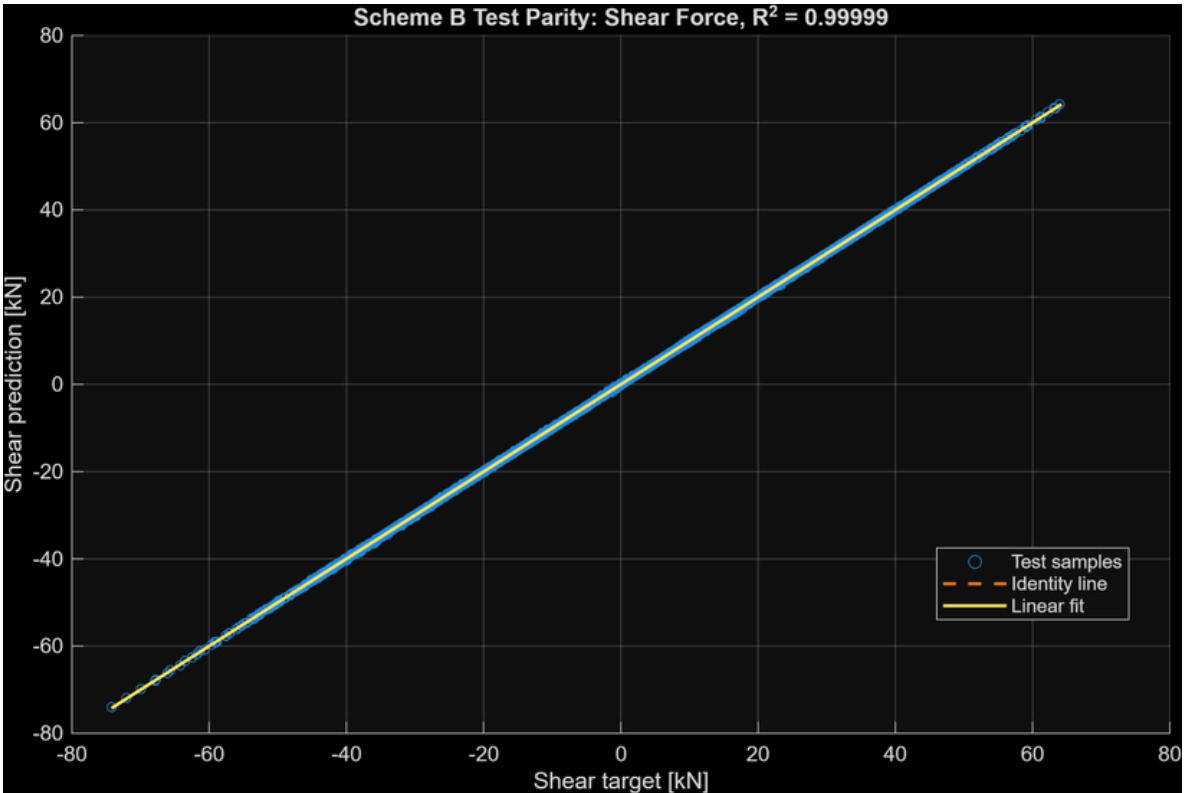
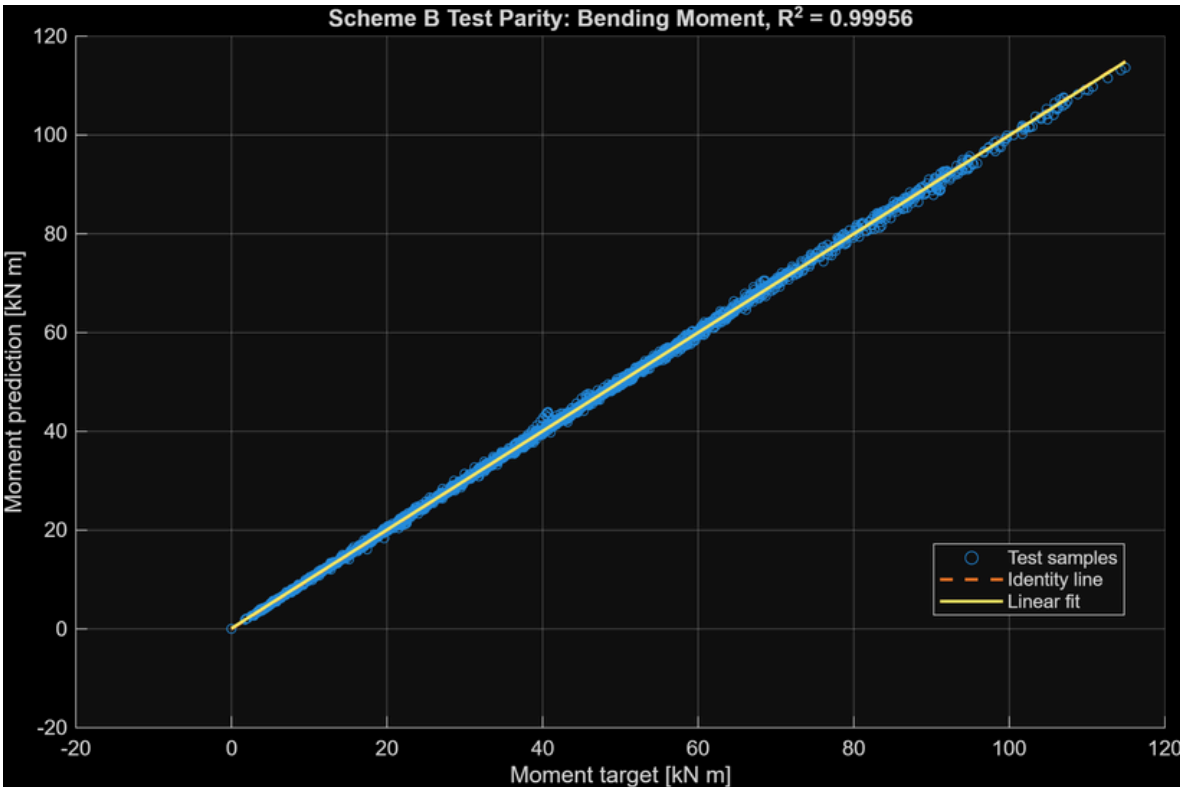
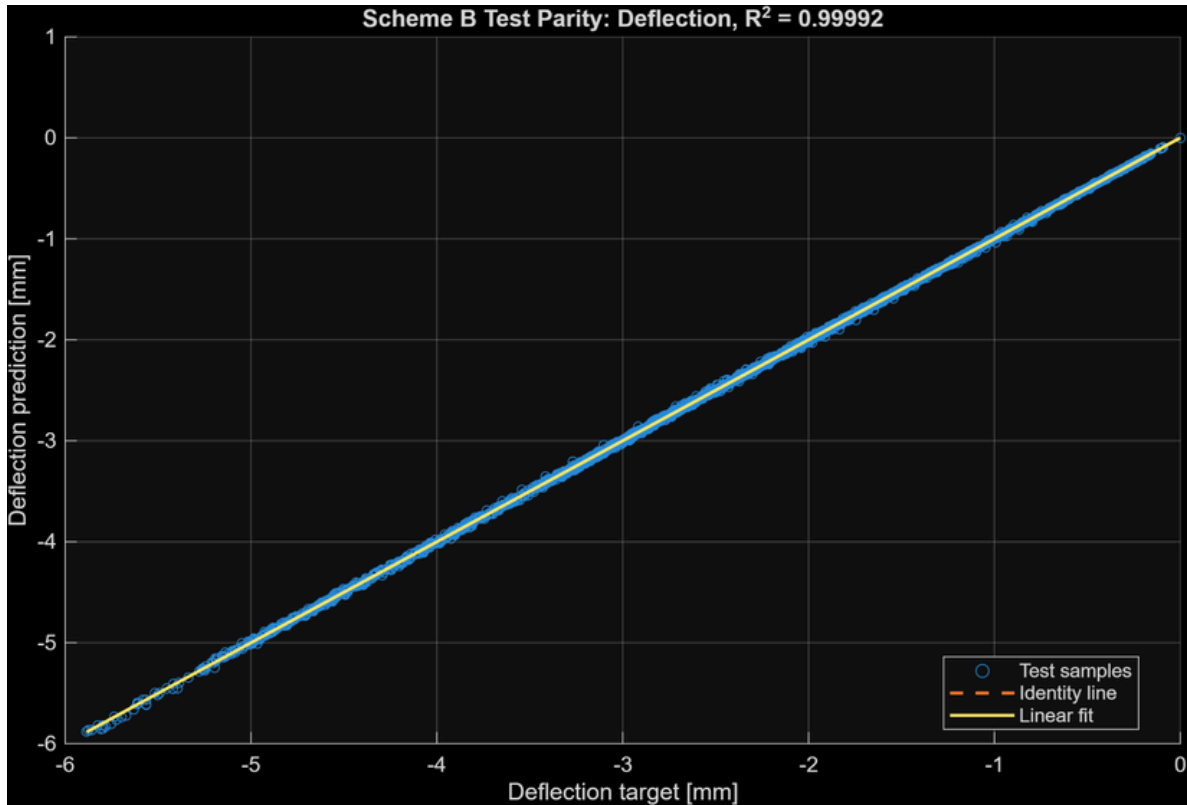
$\xi = x/L$, $\alpha = a/L$, $r = \xi - \alpha$
Inputs: ξ , q/q_{scale} , P/P_{scale} , α , r , $|r|$
Scaled conjugate gradient (trainscg).

What the physics guidance enforces

The factor $\xi(1 - \xi)$ fixes w and M at zero at both supports.
The model measures the shear-jump error but does not enforce it exactly.

Test results

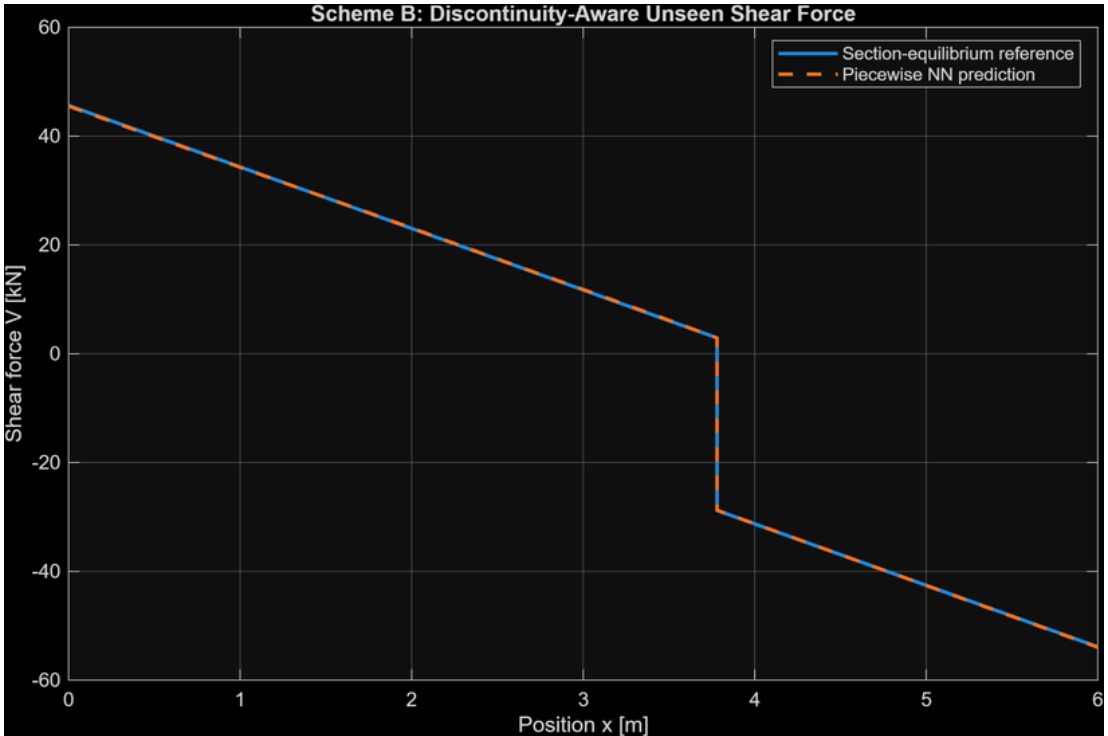
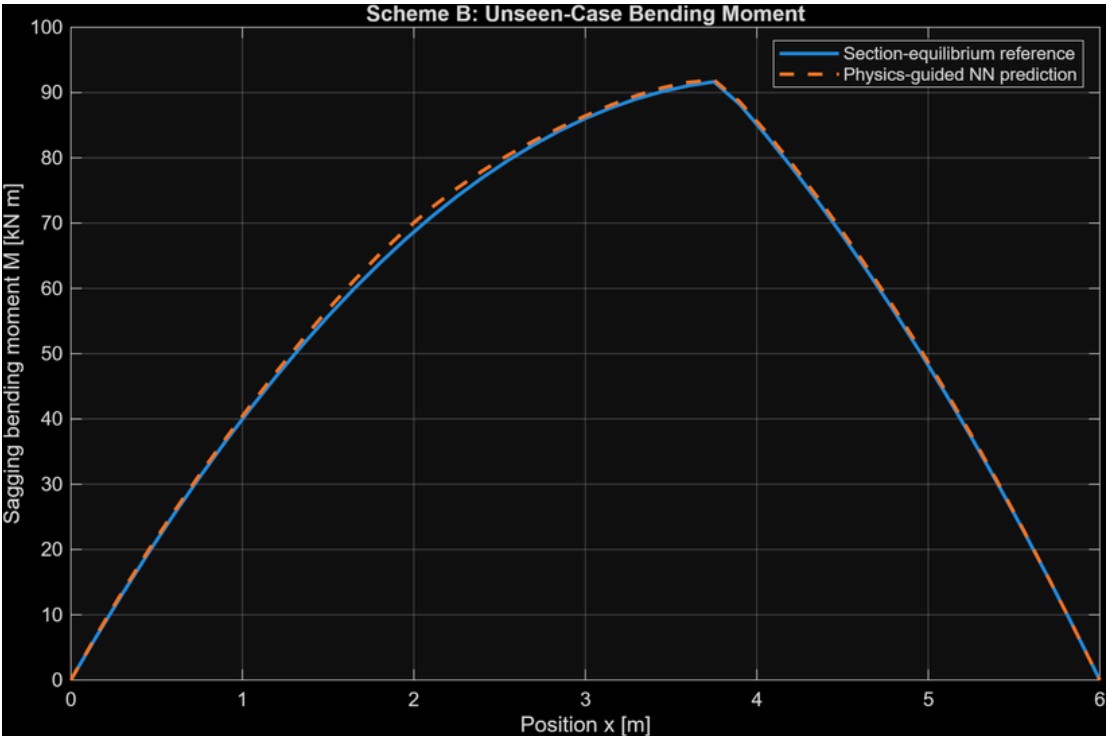
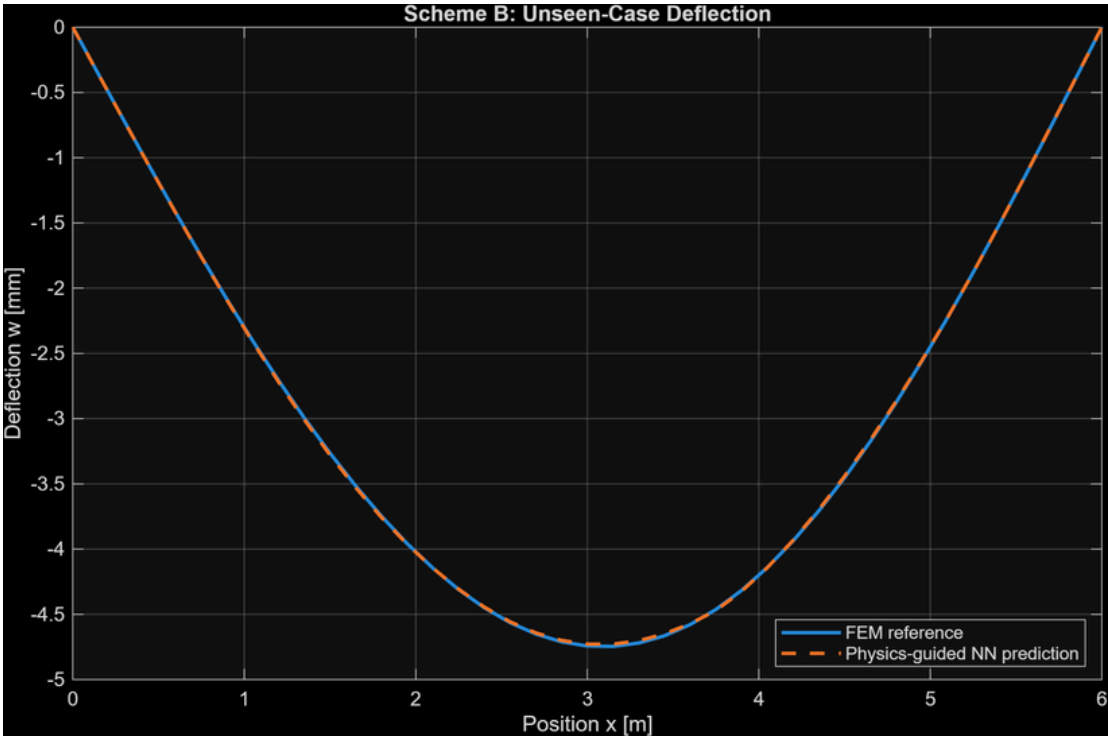
52 complete load cases, separate from training and validation



Response	Test R^2	Test RMSE
Deflection	0.999915	0.013080 mm
Bending moment	0.999559	0.552811 kN·m
Shear force	0.999987	0.099024 kN

Reserved loading case and limitations

$q = 11.3 \text{ kN/m}$, $P = 31.7 \text{ kN}$, $a/L = 0.630$

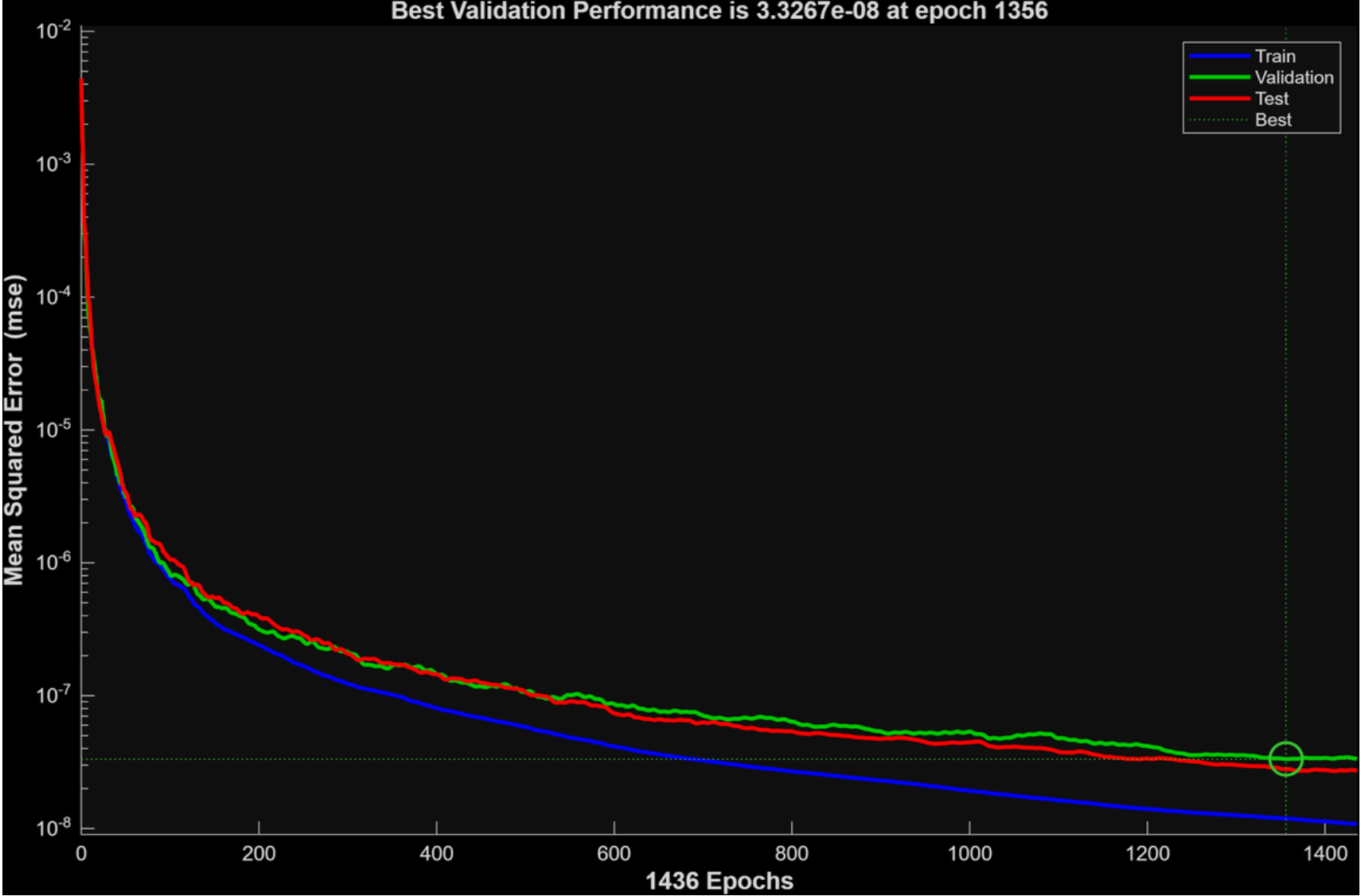


Relative L2 error	Deflection	Moment	Shear
Reserved case	0.247%	1.086%	0.182%

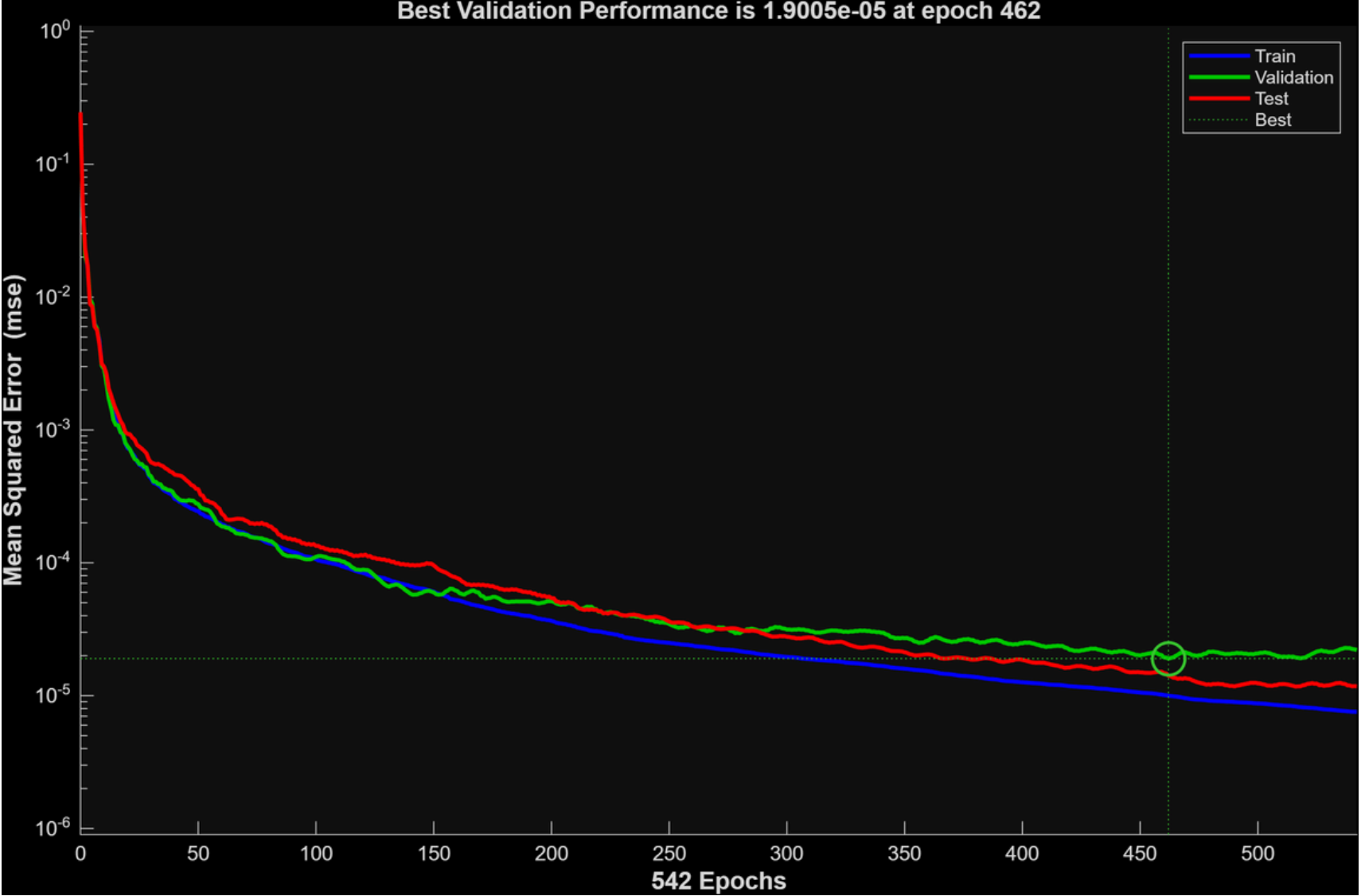
Shear-jump error: 0.084%. Support residuals: 0 mm for deflection and 0 kN·m for moment.

This additional case lies within the sampled loading ranges and was excluded from the 350-case dataset. Evaluation is limited to fixed geometry, material and supports.

Training performance: deflection and bending moment

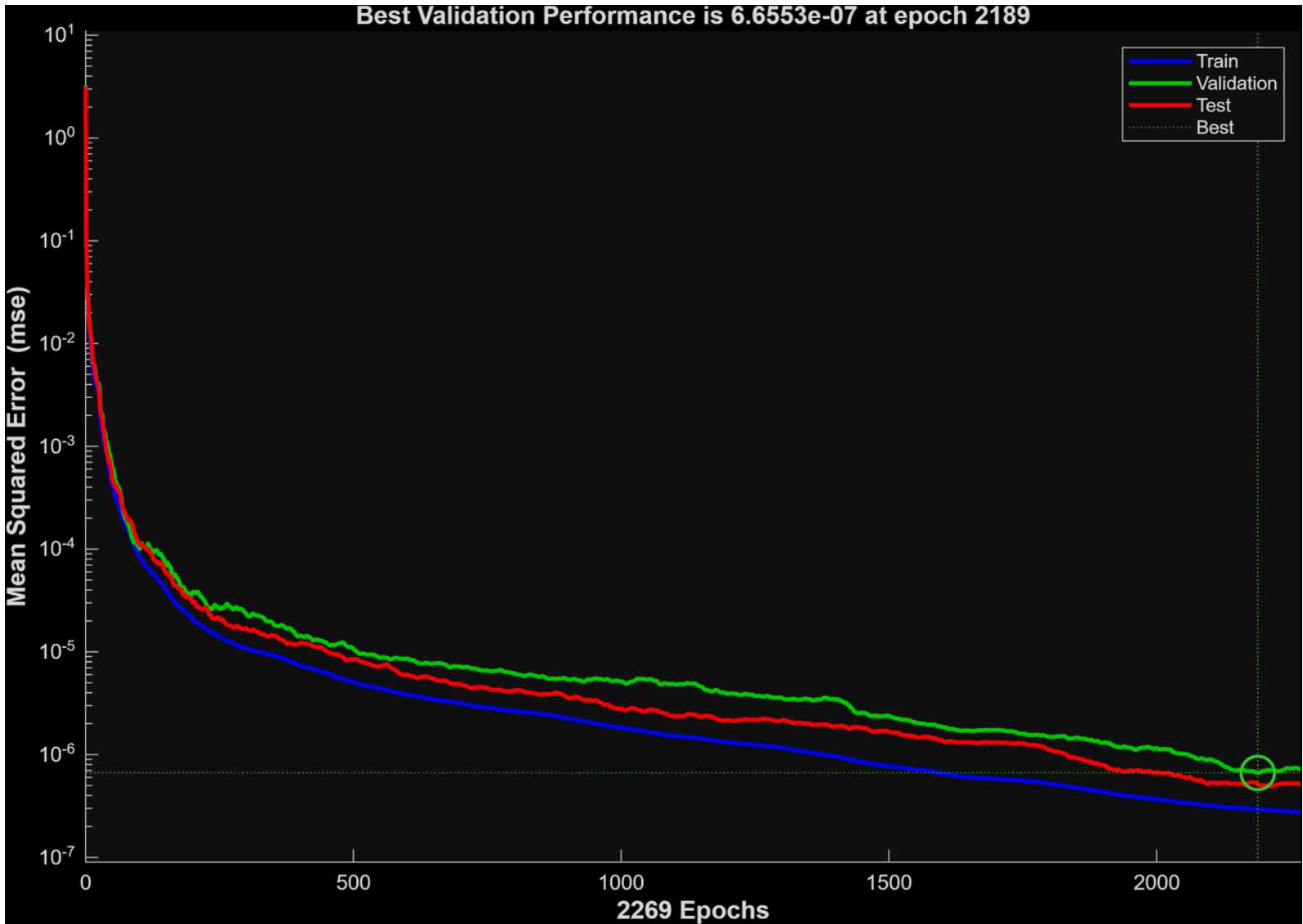


Deflection

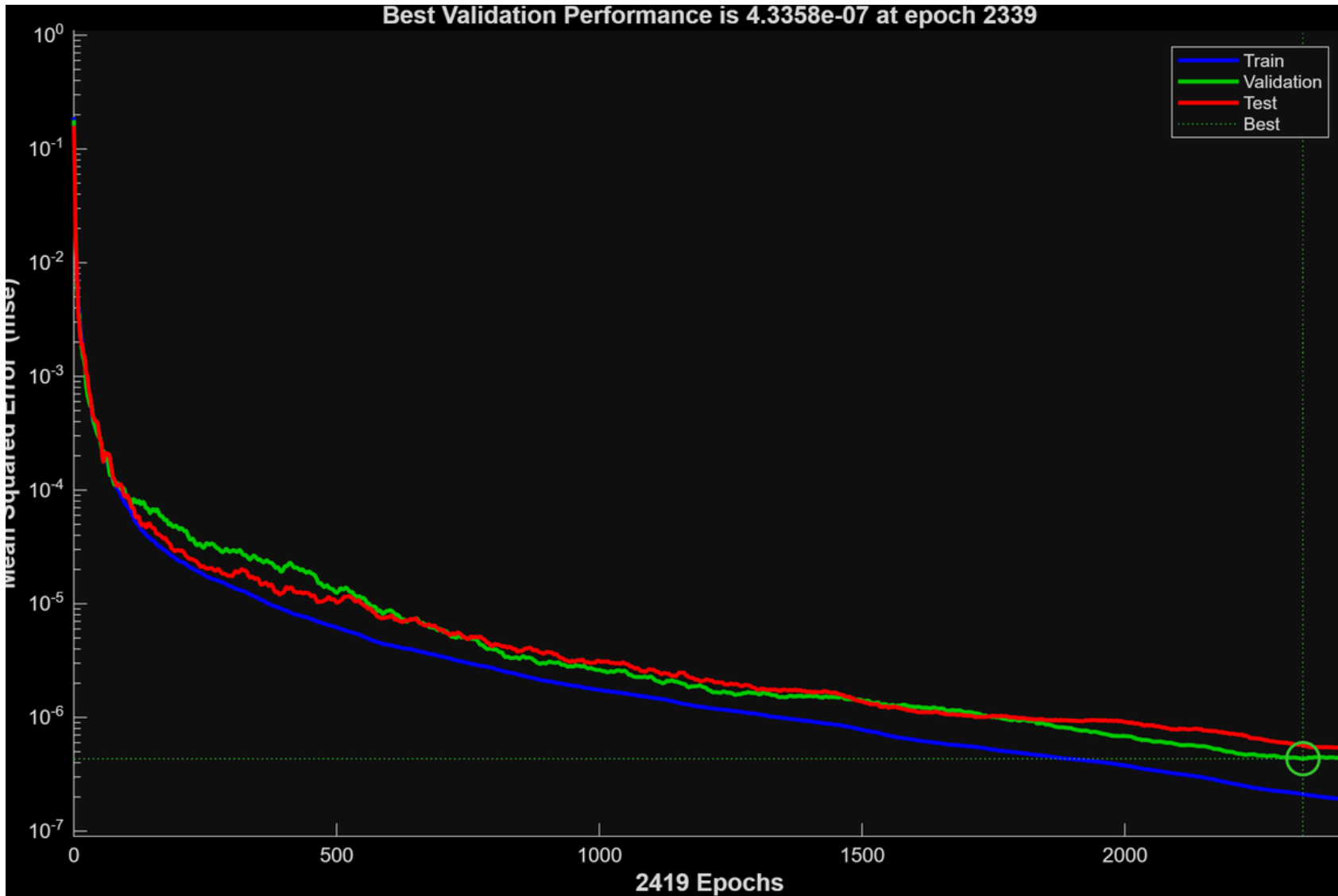


Bending Moment

Training performance: shear force



Left-Side Shear



Right-Side Shear

Early stopping: right side shear network

Best validation epoch: 2339.
Training stopped at epoch 2419 after 80 consecutive epochs without improvement in validation error.
The model from epoch 2339 was retained.

Network Diagram

Training Results
Training finished: Met validation criterion

Training Progress

Unit	Initial Value	Stopped Value	Target Value
Epoch	0	2419	2500
Elapsed Time	-	00:00:45	-
Performance	0.19	1.91e-07	0
Gradient	1.21	6.55e-06	1e-09
Validation Checks	0	80	80

Training Algorithms
Data Division: Index divideind
Training: Scaled Conjugate Gradient trainscg
Performance: Mean Squared Error mse
Calculations: MEX

Training Plots

Performance

Training State

Error Histogram

Regression

Fit